

Topology First-Year Exam, August 2015, Part 1

Answer all questions and work all problems.

1. Suppose that $f : X \rightarrow Y$ is continuous and onto. Show that if X is compact, then Y is also compact.
2. Suppose that $A \subset X$ is connected. Show that the closure of A , $\overline{A}$, is also connected.
3. State and prove the Banach Fixed-Point Theorem.
4. Suppose that X is a compact metric space. Show that if $\{x_i\}$ is a sequence of points in X , then there is a point $z \in X$ and a subsequence, $\{x_{i_j}\}$, such that $x_{i_j} \rightarrow z$.
5. Give an example of a sequence of functions $f_i : [0, 1] \rightarrow [0, 1]$ such that $f_i(x) \rightarrow g(x)$ for all $x \in [0, 1]$, but such that $g : [0, 1] \rightarrow [0, 1]$ is not continuous.
6. Suppose that X is a set and that $f : X \rightarrow 2^X$. Show that f cannot be onto.
7. Suppose that X is a complete metric space that is countably infinite. Show that X has an isolated point, i.e., there is a point $x \in X$ such that $\{x\}$ is open in X .

Answer the following with complete definitions or statements or short proofs.

8. State The Urysohn Lemma.
9. Show that $[0, 1)$ is not compact.
10. State the Tietze Extension Theorem.
11. State the Tychonoff Theorem.
12. State the Alexander Subbase Theorem.
13. Give an example of a Hausdorff space that is not normal.